

Sydney Girls High School



November 2012
MATHEMATICS
EXTENSION 1

YEAR 12
ASSESSMENT TASK 1 for HSC 2013

Time Allowed: 60 minutes
+ 5 minutes Reading Time

Topics: Locus, Integration, Mathematical Induction, Division of an interval in a given ratio, Other Inequalities

General Instructions:

- There are Seven (7) Questions which are not of equal value.
- Attempt all questions.
- Show all necessary working. Marks may be deducted for badly arranged work or incomplete working.
- Start each Question on a new page.
- Write on one side of paper only.
- Diagrams are NOT to scale.
- Board-approved calculators may be used.
- Write your student number clearly at the top of each question and clearly number each question.
- Note that: $\int x^n dx = \frac{1}{n+1}x^{n+1}; \quad n \neq -1$

Total: 50 marks

Student Name: _____

Teacher Name: _____

QUESTION 1 (7 Marks)

MARKS

a) Find a primitive function of each of the following:

i. $5x^3 - 4$ 1

ii. $\frac{7-x^4}{x^2}$ 2

iii. $(8x+3)^5$ 2

b) Find the equation of the parabola, with vertical axis, vertex at the origin and passing through the point (6,3). 2

QUESTION 2 (7 Marks)

MARKS

a) Given the equation of a parabola is $(x+4)^2 = -2(y-8)$, find:

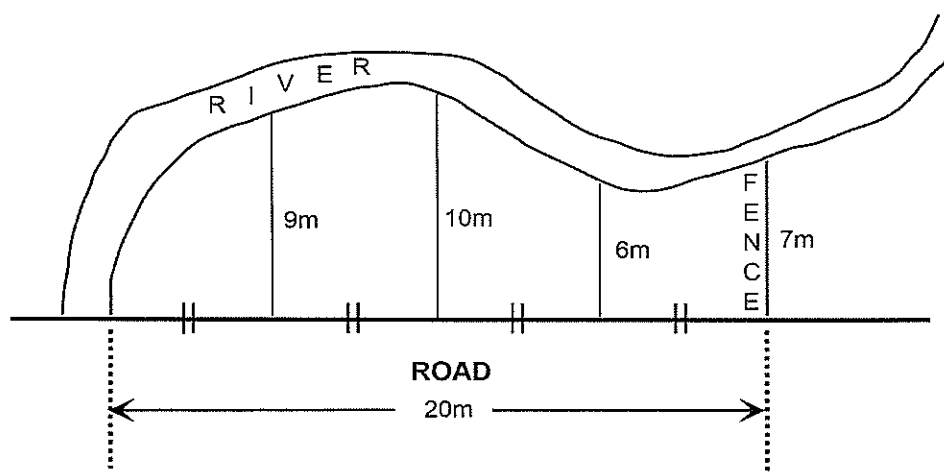
i. the vertex of the parabola. 1

ii. the focal length. 1

b) Find the equation of the parabola with focus (2,0) and directrix $x = -2$. 2

c) A wasteland bordering a river bank and a straight road was fenced off and used as a recreational park. Perpendicular distances from the road to the river bank are shown on the diagram below.

Use Simpson's rule, with 5 function values, to approximate the area of the recreational park. 3



QUESTION 3 (7 Marks)

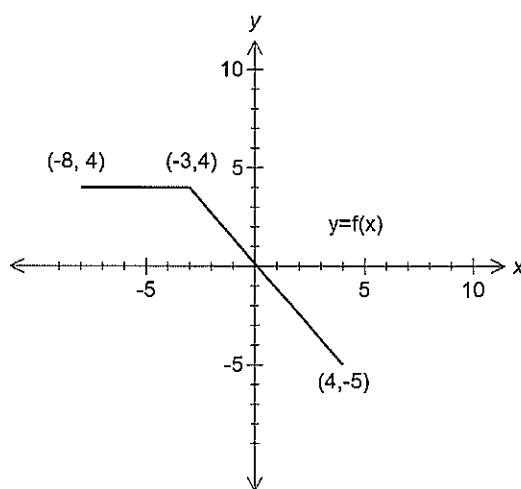
MARKS

- a) Solve the following: $\frac{x}{x-4} \leq -3$. 3
- b) i. Find the x -coordinates of the points of intersection of the line $2x + y = 0$ and the parabola $y = x(x + 4)$. 1
- ii. Find the area bounded by $2x + y = 0$ and $y = x(x + 4)$. 3

QUESTION 4 (7 Marks)

MARKS

- a) Find in what ratio A divides BC, given that the points A(-1,3), B(0,1) and C(1,-1) lie in one line. 3
- b) The point P(k,7) lies on the parabola $x^2 = 20y$.
Find the distance from P to the focus. 2
- c) Given the graph of $y = f(x)$ below, find $\int_{-8}^4 f(x) dx$. 2



QUESTION 5 (7 Marks)

MARKS

- a) i. Draw a diagram showing the area bounded by $y = 2x^3$, $y = 2$ and the y -axis. 1
- ii. The area is now rotated around the x -axis. Calculate the exact volume of the solid of revolution formed. 3
- b) The vertex of a parabola is $V(-1,4)$ and its directrix is $x = 3$.
- i. Find the coordinates of the focus. 1
- ii. Find the equation of the parabola. 1
- iii. Hence, sketch the parabola, showing the main features. 1

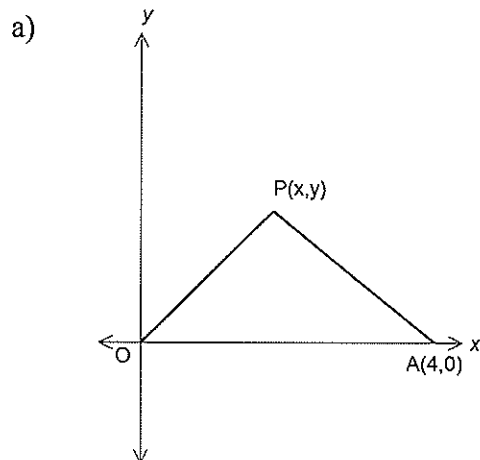
QUESTION 6 (7 Marks)

MARKS

- a) Prove by mathematical induction that $1^2 + 3^2 + \dots + (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$ for $n \geq 1$ 4
- b) Evaluate $\int_{-\frac{8}{3}}^{-1} \frac{2}{\sqrt{1-3x}} dx$ 3

QUESTION 7 (8 Marks)

MARKS



i. A point $P(x, y)$ moves such that $\angle OPA$ is a right angle. Show that the locus of P is given by: $x^2 + y^2 - 4x = 0$. 2

ii. The locus of P is a circle, find its centre and radius. 1

b) A hollow vessel is formed when the parabola $9x^2 = 4y$ is rotated about the y -axis. The distance across the open end of the vessel is 1.5 metres.

i. Show that, when the depth of water in the vessel is h metres, the volume of water is $\frac{2\pi h^2}{9} \text{ m}^3$. 3

ii. Find the depth of the water in the vessel when it is half full. 2

End of Paper

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Extension One Task One: Feedback December 2012

Question	Comment
1	a) For indefinite integrals, don't forget to write "$+C$". b) Students who used a simple diagram and wrote the formula $x^2 = 4ay$ were more successful than those who did not.
2	a) Focal length is $\frac{1}{2}$ not $a = -\frac{1}{2}$ b) Students who used a diagram were more successful than those who did not. Although $x = -2$ was confused with $y = -2$ P.L. c) Tabular Method (or variation) worked better than others
4.	b) k can be found by substitution c) Better to use area formulae rather than calculus
3.	a) Note that $x = 4$ is NOT a solution to the original inequation and must be excluded. b)(ii) Most efficient solution uses $\int (f(x) - g(x)) dx$ where $y = f(x)$ is the upper curve.
5	a) Alternate solution: $V = \text{cylinder} - \text{area under the curve}$. b) Sketching the parabola should include any intercepts; in this case $x = -2$.
6	a) Better solutions factorised the LHS expression in step 3. b) Most common mistake was the integration step (forgetting to divide by -3)
7	b ii) Many students incorrectly interpreted the vessel being half full as meaning the height was $\frac{h}{2}$ rather than the volume being $\frac{V}{2}$, which lead to no marks being awarded.

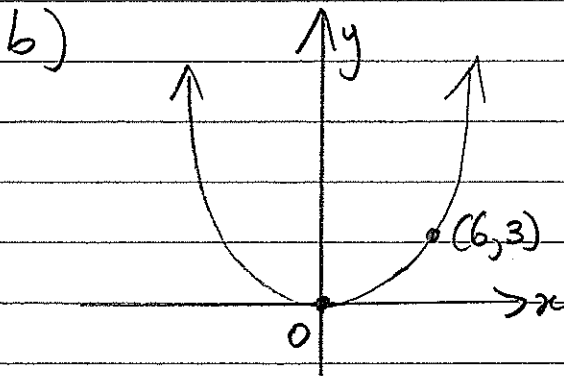
Nov. 2012 Ext. 1 Assessment Task 1 SOLUTIONS

Question 1

a) i) $\int (5x^3 - 4) dx = \frac{5x^4}{4} - 4x + C$ ✓

ii) $\int \left(\frac{7-x^4}{x^2} \right) dx = \int (7x^{-2} - x^2) dx$
 $= \frac{7x^{-1}}{-1} - \frac{x^3}{3} + C$
 $= -\frac{7}{x} - \frac{x^3}{3} + C$ ✓

iii) $\int (8x+3)^5 dx = \frac{(8x+3)^6}{8 \times 6} + C$
 $= \frac{(8x+3)^6}{48} + C$ ✓



$x^2 = 4ay$
Sub in (6,3)

$36 = 4a(3)$

$36 = 12a$

$a = 3$

$\therefore x^2 = 12y$ ✓

Question 2

a) $(x+4)^2 = -2(y-8)$

i) Vertex $(-4, 8)$ ✓

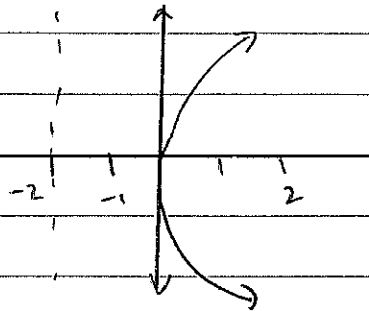
ii) focal length $4a = -2$

$$a = -\frac{1}{2}$$

(2)

i.e. $\frac{1}{2}$ unit ✓

b)



$$a = 2 \quad \checkmark$$

$$y^2 = 4ax$$

$$y^2 = 8x \quad \checkmark$$

(2)

c)

x	$f(x)$	w	$wf(x)$
0	0	1	0
5	9	4	36
10	10	2	20
15	6	4	24
20	7	1	7

$$A \doteq \frac{5}{3} (87)$$

$$= 145 m^2$$

$$\underline{\underline{\sum wf(x) = 87}}$$

(3)

1	2	3	4	5	6	7	8	9	10
1					2	3			

QUESTION 3 (7 Marks)

MARKS

a) Solve the following: $\frac{x}{x-4} \leq -3$.

3

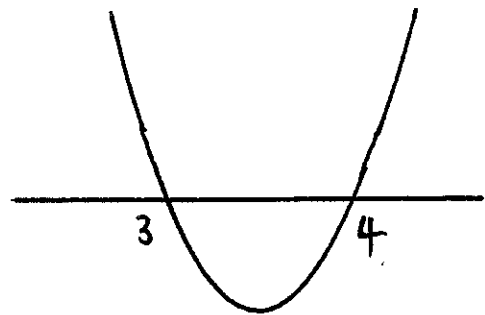
b) i. Find the points of intersection of the line $2x + y = 0$ and the parabola $y = x(x+4)$.

1

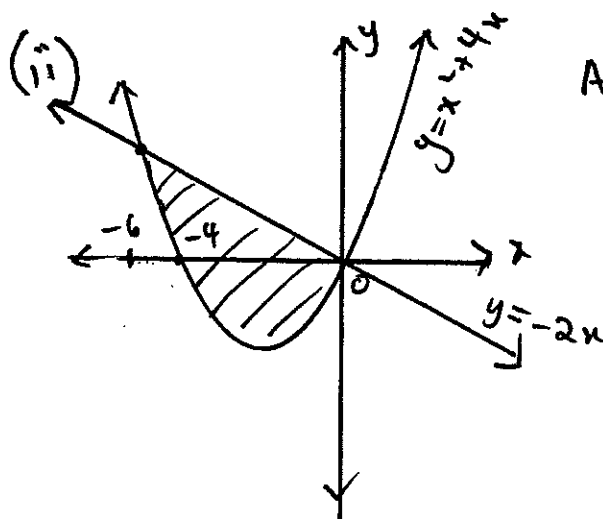
ii. Find the area bounded by $2x + y = 0$ and $y = x(x+4)$.

3

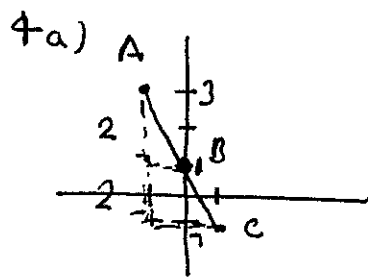
$$\begin{aligned} \text{(a)} \quad x(x-4) &\leq -3(x-4)^2 \\ (x-4)(x+3(x-4)) &\leq 0 \\ (x-4)(4x-12) &\leq 0 \\ \therefore 3 &\leq x \leq 4 \\ \text{but } x &\neq 4 \quad 3 \leq x < 4 \end{aligned}$$



$$\begin{aligned} \text{(b)} \quad \text{(i)} \quad 2x + x(x+4) &= 0 \\ x^2 + 6x &= 0 \quad x(x+6) = 0 \\ \therefore x &= 0 \quad \text{or} \quad x = -6 \end{aligned}$$



$$\begin{aligned} A &= \int_{-6}^0 (-2x - (x^2 + 4x)) dx \\ &= \int_{-6}^0 (-x^2 - 6x) dx \\ &= \left[-\frac{x^3}{3} - 3x^2 \right]_{-6}^0 \\ &= 0 - (72 - 108) \\ A &= 36 \text{ units}^2 \end{aligned}$$



A divides BC
externally 2:1
i.e. 1:2

b) $4a = 20$
 $a = 5$

focus is $(0, 5)$

$$k^2 = 20 \times 7 \quad \checkmark$$

$$= 140$$

$$k = \sqrt{140}$$

$$PS = \sqrt{(\sqrt{140})^2 + (7-5)^2}$$

$$= \sqrt{140 + 4} \quad \checkmark$$

$$= \sqrt{144}$$

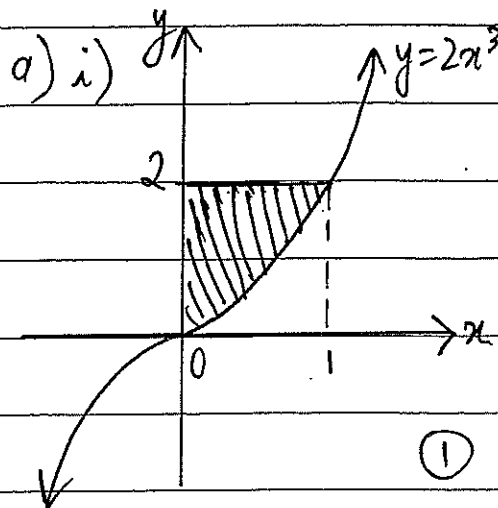
$$= 12$$

c) $\int_{-1}^4 f(x) dx$

$$= 5 \times 4 + \frac{1}{2} \times 3 \times 4 - \frac{1}{2} \times 4 \times 5$$

$$= 16 \quad \checkmark$$

Question 5: (7 Marks)



ii) $V = \pi \int_0^1 y^2 dx$

$$= \pi \int_0^1 2^2 dx - \pi \int_0^1 (2x^3)^2 dx$$

$$= \pi \int_0^1 (4 - 4x^6) dx$$

$$= \pi \left[4x - \frac{4x^7}{7} \right]_0^1$$

$$= \pi \left(4 - \frac{4}{7} \right)$$

$$= \frac{24\pi}{7} \text{ units}^3.$$

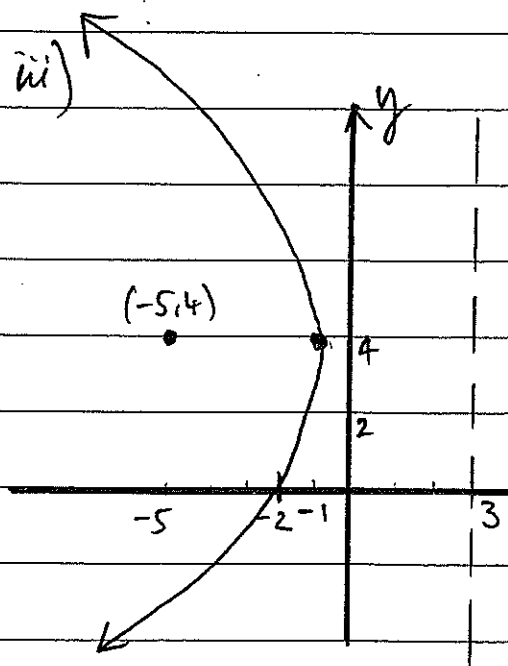
③

b) i) $a = 4$

$$\therefore \text{focus} = (-5, 4) \quad \text{①}$$

$$\text{ii) } (y-h)^2 = -4a(x-k)$$

$$\therefore (y-4)^2 = -16(x+1) \quad \text{①}$$



Question 6:

a) $1^2 + 3^2 + \dots + (2n-1)^2 = \frac{1}{3}n(2n-1)(2n+1)$

Prove true for $n = 1$:

$$LHS = 1^2$$

$$= 1$$

$$RHS = \frac{1}{3} \times 1(2-1)(2+1)$$

$$= 1$$

$$LHS = RHS$$

$\therefore$ Proven true for $n = 1$

Assume true for $n = k$:

$$1^2 + 3^2 + \dots + (2k-1)^2 = \frac{1}{3}k(2k-1)(2k+1)$$

Prove true for $n = k + 1$:

R.T.P $1^2 + 3^2 + \dots + (2k-1)^2 + (2k+1)^2 = \frac{1}{3}(k+1)(2k+1)(2k+3)$

$$LHS = 1^2 + 3^2 + \dots + (2k-1)^2 + (2k+1)^2$$

$$= \frac{1}{3}k(2k-1)(2k+1) + (2k+1)^2 \text{ by assumption}$$

$$= \frac{1}{3}(2k+1)[k(2k-1) + 3(2k+1)]$$

$$= \frac{1}{3}(2k+1)[2k^2 + 5k + 3]$$

$$= \frac{1}{3}(2k+1)(k+1)(2k+3)$$

$$= RHS$$

$\therefore$ proven true for $n = k + 1$

$\therefore$ proven true by mathematical induction for $n \geq 1$

4

b)

$$\int_{-\frac{8}{3}}^{-1} \frac{2}{\sqrt{1-3x}} dx = \int_{-\frac{8}{3}}^{-1} 2(1-3x)^{-\frac{1}{2}} dx$$

$$= 2 \left[\frac{(1-3x)^{\frac{1}{2}}}{-3 \times \frac{1}{2}} \right]_{-\frac{8}{3}}^{-1}$$

$$= -\frac{4}{3} \left[\sqrt{1-3x} \right]_{-\frac{8}{3}}^{-1}$$

$$= -\frac{4}{3} \left(\sqrt{1-3(-1)} - \sqrt{1-3\left(-\frac{8}{3}\right)} \right)$$

$$= -\frac{4}{3} (\sqrt{4} - \sqrt{9})$$

$$= -\frac{4}{3} (2-3)$$

$$= \frac{4}{3}$$

3

Question 7:

a.

$$i. m_{OP} = \frac{y}{x}$$

$$m_{AP} = \frac{y}{x-4}$$

$$m_{OP} \times m_{AP} = -1$$

$$\frac{y}{x} \times \frac{y}{x-4} = -1$$

$$y^2 = -x(x-4)$$

$$y^2 + x^2 - 4x = 0$$

/2

$$ii. y^2 + x^2 - 4x + 4 = 4$$

$$y^2 + (x-2)^2 = 4$$

$$\text{centre } (2,0) \quad \text{radius} = 2$$

/1

b.

$$i. V = \pi \int_0^h x^2 dy$$

$$= \pi \int_0^h \frac{4y}{9} dy$$

$$= \pi \left[\frac{2y^2}{9} \right]_0^h$$

$$= \frac{2\pi h^2}{9}$$

/3

ii. When vessel is full:

$$x = \frac{3}{4}$$

$$y = \frac{9}{4} \times \left(\frac{3}{4} \right)^2$$

$$= \frac{81}{64}$$

$$V = \frac{2\pi}{9} \times \left(\frac{81}{64} \right)^2$$

$$= \frac{729\pi}{2048}$$

When vessel is half full:

$$V = \frac{729\pi}{4096}$$

$$\frac{729\pi}{4096} = \frac{2\pi h^2}{9}$$

$$h^2 = \frac{6561}{8192}$$

$$h = 0.89$$

/2